F12 4th CAR

# Lecture 32: Section 5.1
# Fundamental Trigonometric Identities

Reciprocal identities

Quotient identities

Pythagorean identities

Even/odd identities

Cofunction identities

Simplify trigonometric expressions

Factor trigonometric expressions

Add trigonometrice expressions

Trigonometric substitution

Simplify logarithmic expressions

# Lecture 32: Section 5.1
# Fundamental Trigonometric Identities

1. Reciprocal Identities:

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

2. Quotient Identities:

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

3. Pythagorean Identities:

$$\sin^2\theta + \cos^2\theta = 1$$

$f(-x) = f(x)$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

$f(-x) = -f(x)$

4. Even/Odd Identities:

odd $\sin(-\theta) = -\sin\theta$ $\qquad$ $\cos(-\theta) = \cos\theta$ even

odd $\tan(-\theta) = -\tan\theta$ $\qquad$ $\cot(-\theta) = -\cot\theta$ odd

even $\sec(-\theta) = \sec\theta$ $\qquad$ $\csc(-\theta) = -\csc\theta$ odd

$\sin(90-30)^\circ = \cos 30^\circ$

$\sin 60^\circ = \cos 30^\circ$

5. Cofunction Identities:

$$\sin\left(\frac{\pi}{2}-\theta\right) = \cos\theta \qquad \cos\left(\frac{\pi}{2}-\theta\right) = \sin\theta$$

$$\tan\left(\frac{\pi}{2}-\theta\right) = \cot\theta \qquad \cot\left(\frac{\pi}{2}-\theta\right) = \tan\theta$$

$$\sec\left(\frac{\pi}{2}-\theta\right) = \csc\theta \qquad \csc\left(\frac{\pi}{2}-\theta\right) = \sec\theta$$

odd

$-\sin\theta$

thus

$\frac{\sin\theta}{\cos\theta} = \frac{+}{-}$

**ex.** If $\sin(-\theta) = -\frac{2}{3}$ and $\tan\theta < 0$, evaluate all six trigonometric functions of $\theta$.

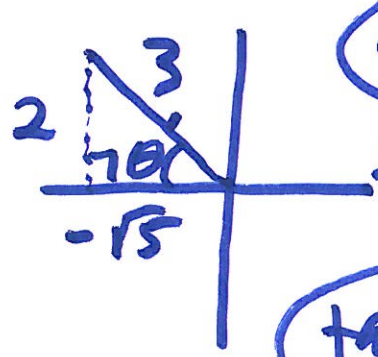
II or III

$-\sin\theta = -\frac{2}{3}$

Method 1:

$\sin\theta = \frac{2}{3}$ quad I or II

$\csc\theta = \frac{3}{2}$




$\cos\theta = -\frac{\sqrt{5}}{3}$

$\tan\theta = \frac{2}{-\sqrt{5}} =$

$\tan\theta = -\frac{2\sqrt{5}}{5}$

$\sec\theta = -\frac{3}{\sqrt{5}} = -\frac{3\sqrt{5}}{5}$

$\cot\theta = -\frac{\sqrt{5}}{2}$

Method 2: $\sin\theta = \frac{2}{3}$ quad II

$$\sin^2\theta + \cos^2\theta = 1$$

$$\left(\frac{2}{3}\right)^2 + \cos^2\theta = 1$$

$$\cos^2\theta = \frac{5}{9}$$

$$\cos\theta = \pm\frac{\sqrt{5}}{3} = -\frac{\sqrt{5}}{3} = \cos\theta$$

$$\csc\theta = \frac{3}{2}$$

$$\sec\theta = -\frac{3}{\sqrt{5}} = -\frac{3\sqrt{5}}{5}$$

$$\tan\theta = \frac{\frac{2}{3}}{-\frac{\sqrt{5}}{3}} = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}$$

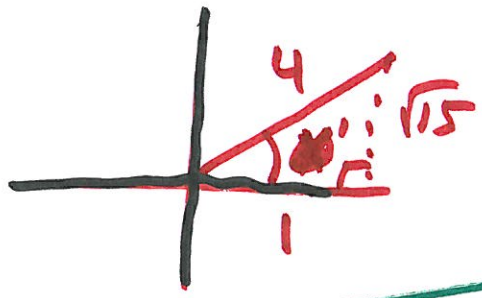
$$\cot\theta = -\frac{\sqrt{5}}{2}$$

(+,+) thus quad I

**ex.** If $\csc\left(\frac{\pi}{2} - x\right) = 4$ and $\sin x > 0$, evaluate all six trigonometric functions of x.

$\csc\left(\frac{\pi}{2} - x\right) = 4 = \sec x$ thus $\cos x = \frac{1}{4}$

Method 1:




$1^2 + a^2 = 4^2$

$a^2 = 15$

$a = \sqrt{15}$

$\sin x = \frac{\sqrt{15}}{4}$

$\csc x = \frac{4}{\sqrt{15}} = \frac{4\sqrt{15}}{15}$

$\tan\theta = \frac{\frac{\sqrt{15}}{4}}{\frac{1}{4}} = \sqrt{15}$

$\cot\theta = \frac{1}{\sqrt{15}} = \frac{\sqrt{15}}{15}$

Method 2:

$\csc\left(\frac{\pi}{2} - x\right) = 4 = \sec x$

$\cos x = \frac{1}{4}$

$\sin^2 x + \left(\frac{1}{4}\right)^2 = 1$

$\sin^2 x = \frac{15}{16}$

$\sin x = \pm \frac{\sqrt{15}}{4} = +\frac{\sqrt{15}}{4}$

etc.

(1) Look for identities.

(2) Change to sine and cosine.

**ex.** Simplify:

1) $\cot^2\theta(\overbrace{1+\tan^2\theta}^{\sec^2\theta}) = \cot^2\theta \cdot \sec^2\theta =$

$$\frac{\cancel{\cos^2\theta}}{\sin^2\theta} \cdot \frac{1}{\cancel{\cos^2\theta}} = \frac{1}{\sin^2\theta}$$

$$= \boxed{\csc^2\theta}$$

2) $\dfrac{\tan(-\theta)}{\sec\theta} = \dfrac{-\tan\theta}{\sec\theta} = -\dfrac{\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos\theta}} =$

$$= -\frac{\sin\theta}{\cancel{\cos\theta}} \cdot \frac{\cancel{\cos\theta}}{1} = -\sin\theta$$

or $= \sin(-\theta)$

3) $\dfrac{\overbrace{\sec^2\theta - 1}^{\tan^2\theta}}{\sin^2\theta} = \dfrac{\tan^2\theta}{\sin^2\theta} = \dfrac{\frac{\sin^2\theta}{\cos^2\theta}}{\frac{\sin^2\theta}{1}} =$

$$= \frac{\cancel{\sin^2\theta}}{\cos^2\theta} \cdot \frac{1}{\cancel{\sin^2\theta}} = \frac{1}{\cos^2\theta} = \sec^2\theta$$

**ex.** Perform the operations and simplify:

1) $(1+\tan x)^2 - 2\tan x =$

$$1 + \cancel{2\tan x} + \tan^2 x - \cancel{2\tan x} =$$

$$1 + \tan^2 x = \boxed{\sec^2 x}$$

2) $\sin t + \cot t \cos t =$ $\sin t + \dfrac{\overbrace{\cos t}^{\cos^2 t}}{\sin t} \cdot \cos t =$

$$= \left(\frac{\sin t}{\sin t}\right)\sin t + \frac{\cos^2 t}{\sin t} = \frac{\sin^2 t + \cos^2 t}{\sin t} = \frac{1}{\sin t} = \boxed{\csc t}$$

3) $\left(\dfrac{1+\sin\theta}{1+\sin\theta}\right)\dfrac{\sin\theta}{\cos\theta} + \left(\dfrac{\cos\theta}{1+\sin\theta}\right)\left(\dfrac{\cos\theta}{\cos\theta}\right) = \dfrac{\sin\theta + \overbrace{\sin^2\theta + \cos^2\theta}^{1}}{\cos\theta(1+\sin\theta)} =$

$$= \frac{\cancel{(\sin\theta + 1)}}{\cos\theta\cancel{(\sin\theta + 1)}} = \frac{1}{\cos\theta} = \boxed{\sec\theta}$$

mult. by conjugate

**ex.** Rewrite $\dfrac{1}{1-\sin x}$ so that it is not in fractional form.

$$\left(\frac{1}{1-\sin x}\right)\left(\frac{1+\sin x}{1+\sin x}\right) = \frac{1+\sin x}{1-\sin^2 x} = \frac{1+\sin x}{\cos^2 x} =$$

$$\frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} = \sec^2 x + \frac{\sin x}{\cos x}\cdot\frac{1}{\cos x} =$$

$$= \boxed{\sec^2 x + \tan x \sec x}$$

$\sin^2\theta + \cos^2\theta = 1$

**NOTE:** From Pythagorean identities, we have

$$\begin{aligned}
\sin^2\theta &= 1-\cos^2\theta = (1+\cos\theta)(1-\cos\theta)\\
\cos^2\theta &= 1-\sin^2\theta = (1+\sin\theta)(1-\sin\theta)\\
\tan^2\theta &= \sec^2\theta - 1 = (\sec\theta+1)(\sec\theta-1)\\
\cot^2\theta &= \csc^2\theta - 1 = (\csc\theta+1)(\csc\theta-1)\\
1 &= \sec^2\theta - \tan^2\theta\\
&= (\sec\theta+\tan\theta)(\sec\theta-\tan\theta)\\
1 &= \csc^2\theta - \cot^2\theta\\
&= (\csc\theta+\cot\theta)(\csc\theta-\cot\theta)
\end{aligned}$$

$\tan^2\theta + 1 = \sec^2\theta$

$\cot^2\theta + 1 = \csc^2\theta$

$u^3 - v^3 = (u-v)(u^2+uv+v^2)$

**ex.** Factor:

1) $\sin^3\theta - \cos^3\theta = (\sin\theta - \cos\theta)(\sin^2\theta + \sin\theta\cos\theta + \cos^2\theta)$

$= \boxed{(\sin\theta - \cos\theta)(1 + \sin\theta\cos\theta)}$

$1+\tan^2 x$

2) $2\sec^2 x + 3\tan x - 1 =$

$2(1+\tan^2 x) + 3\tan x - 1 =$

$2 + 2\tan^2 x + 3\tan x - 1 =$

$2\tan^2 x + 3\tan x + 1 =$

$\boxed{(2\tan x + 1)(\tan x + 1)}$

## Trigonometric Substitution

(+,+)

<u>ex.</u> Use the substitution $x = 2\sin\theta$, $0 < \theta < \frac{\pi}{2}$ to write $\frac{x}{\sqrt{4-x^2}}$ as a trigonometric function of $\theta$.

Method 1:

$$\frac{2\sin\theta}{\sqrt{4-(2\sin\theta)^2}} = \frac{2\sin\theta}{\sqrt{4-4\sin^2\theta}} = \frac{2\sin\theta}{\sqrt{4(1-\sin^2\theta)}} =$$

$$= \frac{2\sin\theta}{2\sqrt{1-\sin^2\theta}} = \frac{2\sin\theta}{2\sqrt{\cos^2\theta}} = \frac{\cancel{2}\sin\theta}{\cancel{2}\cos\theta} = \boxed{\tan\theta}$$

<u>ex.</u> Use the substitution $x = \frac{5}{2}\tan\theta$, $0 < \theta < \frac{\pi}{2}$ to rewrite the equation $\sqrt{4x^2+25} = 10$ and then find $\sec\theta$ and $\tan\theta$.

(+,+)

$$\sqrt{4\left(\tfrac{5}{2}\tan\theta\right)^2+25} = 10$$

$$\sqrt{\cancel{4}\left(\tfrac{25}{\cancel{4}}\tan^2\theta\right)+25} = 10$$

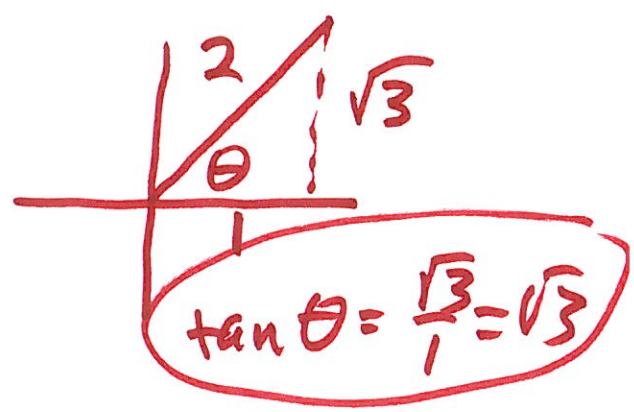
$$\sqrt{25(\tan^2\theta+1)} = 10$$

$$5\sqrt{\sec^2\theta} = 10$$

$$5\sec\theta = 10$$

$$\sec\theta = 2$$

thus $\frac{\text{hyp}}{\text{adj}} = \sec\theta = \frac{2}{1}$




$$\tan\theta = \frac{\sqrt{3}}{1} = \sqrt{3}$$

## Simplify Logarithmic Expressions

**ex.** Rewrite the logarithmic expression

$$\ln|\sec\theta| + \ln|\cot\theta| \quad =$$

as a single logarithm and simplify the result.

$$= \ln|\sec\theta \cdot \cot\theta| = \ln\left|\frac{1}{\cancel{\cos\theta}} \cdot \frac{\cancel{\cos\theta}}{\sin\theta}\right| = \ln\left|\frac{1}{\sin\theta}\right|$$

$$= \ln|\csc\theta|$$

**ex.** Rewrite the logarithmic expression

$$\ln(1 - \sin^2 x) - \ln(\csc^2 x - 1) \quad =$$

as a single logarithm and simplify the result.

$$= \ln\left(\frac{1-\sin^2 x}{\csc^2 x - 1}\right) = \ln\frac{\cos^2 x}{\cot^2 x} = \ln\frac{\cos^2 x}{\frac{\cos^2 x}{\sin^2 x}} =$$

$$= \ln \cancel{\cos^2 x} \cdot \frac{\sin^2 x}{\cancel{\cos^2 x}} = \ln \sin^2 x = \ln(\sin x)^2$$

$$= \boxed{2\ln|\sin x|}$$



$\ln y^2 = 2\ln y$